IITM PRAVARTAK
CATALYSING INNOVATION

Professional Certificate Programme in Agentic AI and Applications

Decision Making and Planning in Agents

Revisit the Learning So Far

Let’s take a quick look at the key ideas and concepts you have explored in your learning journey.

- How agentic AI system architectures are designed and structured
- Fundamental types of AI agents and how they operate across industry use cases
- The differences between task-oriented and goal-oriented agents and how agents communicate within coordinated frameworks

Topics Covered

- Goal-Setting in Autonomous Agent
- Real–World Applications
- Hierarchical v. Reactive Planning
- Planning Strategies in Action
- Autonomous Agents
- Multi-Agent Planning Strategies
- Reinforcement Loops in AI Planning

Goal-Setting in Autonomous Agents

The strategic framework that transforms AI instructions into actionable intelligence

Defining Goal-Setting in Autonomous Systems

Goal-setting transforms reactive systems into proactive, strategic decision-makers capable of independent operation.

- Represents a fundamental cognitive process for interpreting instructions or environmental needs into structured, actionable goals

This capability distinguishes autonomous agents from simple function-calling systems.

- Relies on decision-making algorithms to understand intent, prioritise actions and define success criteria.
- Enables agents to navigate uncertainty, allocate resources and adapt to changing conditions.

The Strategic Importance of Goal Architecture

Agents use clear goals to evaluate competing objectives and allocate computational resources effectively.

Agents apply goal-driven reasoning to select appropriate APIs and methodologies for specific tasks.

Agents define quantifiable goals to assess performance, refine strategies and report meaningful feedback.

Task prioritisation

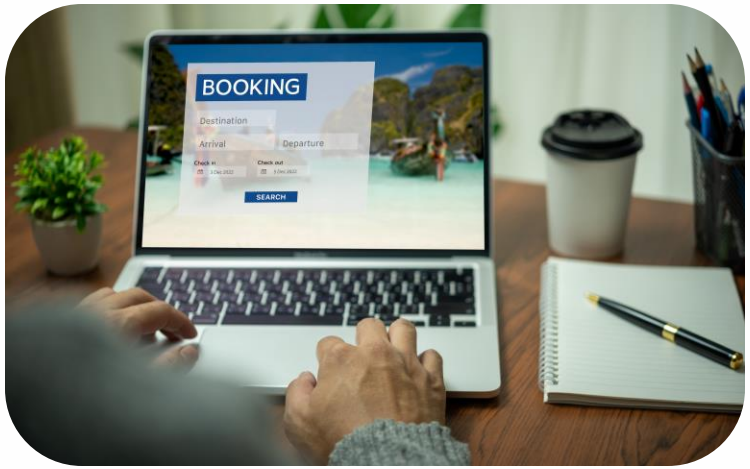
Tool selection

Success measurement

Three Fundamental Goal Categories

Human-defined goals

- Agents execute direct instructions from users or administrators, interpreting explicit objectives with precision.
- **Example**: “Summarise today’s financial news and send a 1-page report” — a clear, bounded task with specific deliverables.



System-generated goals

- Agents decompose complex tasks into structured sub-objectives through intelligent planning.
- **Example**: Trip planning breaks into flight search → price comparison → booking → confirmation storage, each with distinct success metrics.

Adaptive goals

- Agents adjust objectives dynamically in response to environmental changes and real-time feedback.
- **Example**: A warehouse robot maintains “deliver item” while rerouting around obstacles or traffic patterns.

The Goal Execution Pipeline

Interpretation: Agents analyse inputs to identify intent, context and constraints through semantic parsing and disambiguation.

Decomposition: Agents break complex goals into structured sub-tasks with dependencies and success criteria for parallel execution and error isolation.

Planning: Agents design action sequences, select appropriate tools and define logic for contingencies and failure handling.

Execution: Agents carry out planned actions via API calls, data processing and system interactions while tracking progress.

Reflection: Agents evaluate outcomes against success criteria, refine strategies and enable learning for future tasks.

Real-World Applications

Professional autonomous agents in production environments

Customer Support Agent Architecture

Primary Goal Framework

Core objective: Resolve user queries with maximum efficiency and satisfaction

1. **Issue identification:** Parse input, classify problem and assess urgency via NLP and pattern recognition.
2. **Knowledge base search:** Search internal documentation and solution repositories via semantic matching and relevance algorithms.
3. **Solution generation:** Synthesise findings into actionable responses, adjusting tone to match user expertise.
4. **Escalation logic:** Initiate human handoff when complexity exceeds system limits or satisfaction declines.

Financial Research Agent Implementation

Example goal: Deliver Tesla investment analysis with volatility, regulation, and competition insights within 15 minutes.

- **Data acquisition:** Fetch live market data, news and filings from financial APIs with error handling for unavailable sources.
- **Analysis engine:** Process quantitative metrics, news sentiment, and technical indicators to generate insights.
- **Report generation:** Synthesise findings into investment reports with risks, targets and recommendations.
- **Adaptive fallback:** Switch to cached data when APIs fail to maintain report generation.

Healthcare Assistant Critical Decision Making

Mission-Critical Goal Hierarchy

Primary objective: Support clinical decision-making through intelligent patient triage and risk assessment

Dynamic goal adaptation triggers emergency alerts when vitals signal danger, bypassing standard triage.

Data integration
Combine patient history, current symptoms, and vitals from health records and monitors.

Risk stratification
Apply clinical algorithms and patterns to assess severity and set priorities.

Emergency override
Escalate life-threatening cases instantly, bypassing standard workflows.

Key Takeaways for Professional Implementation

Autonomy through goals: Goal-setting enables autonomous agents to plan and decide independently, unlike function-calling systems.

Hierarchical structure: Successful agents decompose complex goals into sub-tasks with dependencies and success metrics.

Dynamic adaptation: Agents adjust goals in real time to stay effective amid changing conditions.

Performance measurement: Robust goal frameworks include measurable criteria and feedback for ongoing optimisation.

Governance framework: Human oversight ensures goals align with ethics, policy and regulation.

Real-World Impact: AI Patient Triage

- AI triage assistants merge autonomous intelligence with urgent healthcare delivery.
- Sophisticated goal-setting enables autonomous agents to operate in high-stakes clinical environments.
- Effective triage showcases five goal-setting dimensions: clarity, hierarchy, adaptability, measurability and governance.

Foundation: High-Level Goal Definition

Primary objective

Help doctors triage patients efficiently in emergency departments

- Process patient information systematically
- Assess medical urgency using standardized protocols
- Suggest appropriate priority levels for treatment

Operational translation

Transform abstract healthcare goals into executable agent behaviours

- Collect comprehensive patient data
- Apply clinical decision-making algorithms
- Generate actionable triage recommendations

Effective goal-setting begins with a clear definition of the agent's purpose, guiding flexible execution.

Hierarchical Task Decomposition

Retrieve patient history
Access health records to review medical background, conditions, and treatments.

Collect real-time vitals
Gather current measurements: heart rate, blood pressure, temperature and oxygen levels.

Analyse symptom reports
Process reported symptoms and pain levels from forms and assessments.

Compare against guidelines
Match findings to medical protocols and severity indices.

Generate triage recommendation
Classify priority as Critical, Urgent or Routine with supporting rationale.

Dynamic Adaptation in Critical Scenarios

Vital sign deterioration

The agent escalates priority and activates critical care when vitals drop, bypassing standard triage.

Missing data handling

The system uses intake data and vitals when records are unavailable.

Multi-patient prioritisation

The agent reorders cases by severity during high-volume periods, not arrival time.

Dynamic Adaptation in Critical Scenarios

Real-world healthcare environments require agents that adapt instantly to changing conditions.

The triage system maintains effectiveness despite incomplete data, system failures or patient surges.

Comprehensive Success Metrics

94% diagnostic accuracy
AI triage matches physician assessments 94% of the time.

<15s response timeliness
Triage recommendations are generated within a maximum of 15 seconds of data input.

99.2% system reliability
Uptime and graceful handling of missing or corrupted patient data.

–23% wait time reduction
Cuts average wait times for critical and urgent cases by 23%.

These benchmarks objectively assess agent performance and support data-driven triage optimisation.

Governance and Ethical Boundaries

Human-in-the-loop design
Physicians retain final authority; the AI offers recommendations, not autonomous decisions.

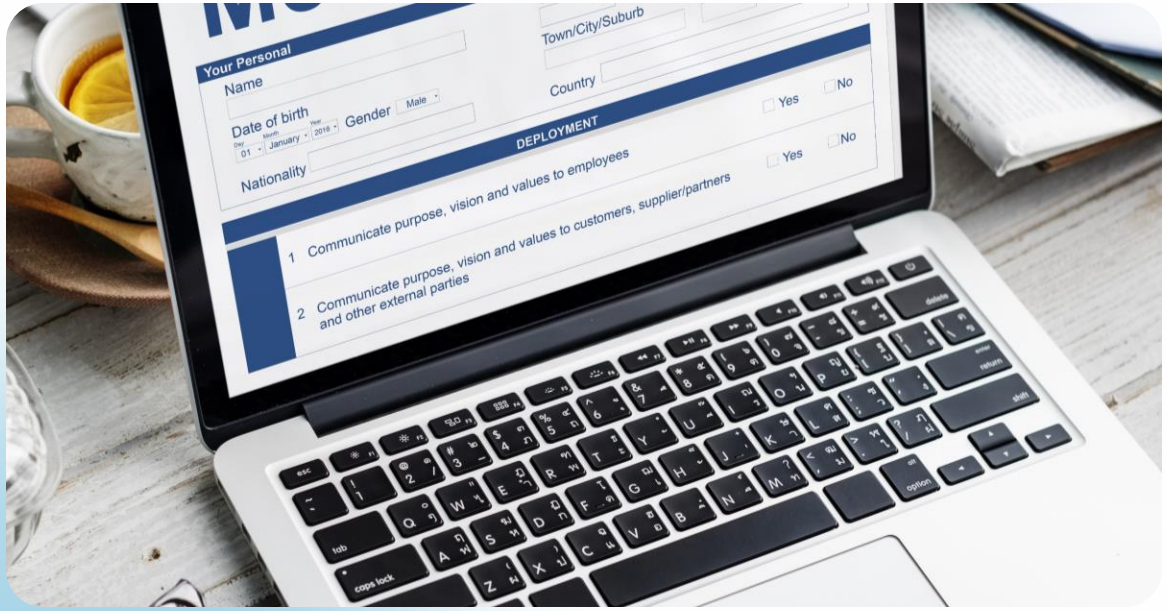
Complete audit trail
Every agent recommendation is logged with timestamps, inputs, reasoning and outcomes for compliance and quality assurance.

Bias prevention
Algorithms block demographic-based bias, ensuring fair care across age, gender, race and socioeconomic status.

HIPAA compliance
Patient data is handled securely with encryption, access controls, and retention policies that meet privacy standards.

Integration Architecture

The triage assistant connects smoothly to hospital systems, linking records, monitors and nurse stations.



Real-time data keeps the agent aware of current conditions while meeting security and workflow standards.

- EHR system integration for patient history
- IoT device connectivity for live vitals
- Nurse station dashboard integration
- Hospital communication system alerts

Framework Summary

Goal setting: Defines what success looks like through clear, measurable objectives

Hierarchy: Structures complex tasks into manageable, sequential components

Dynamic adaptation: Demonstrates real-world resilience when conditions change unexpectedly

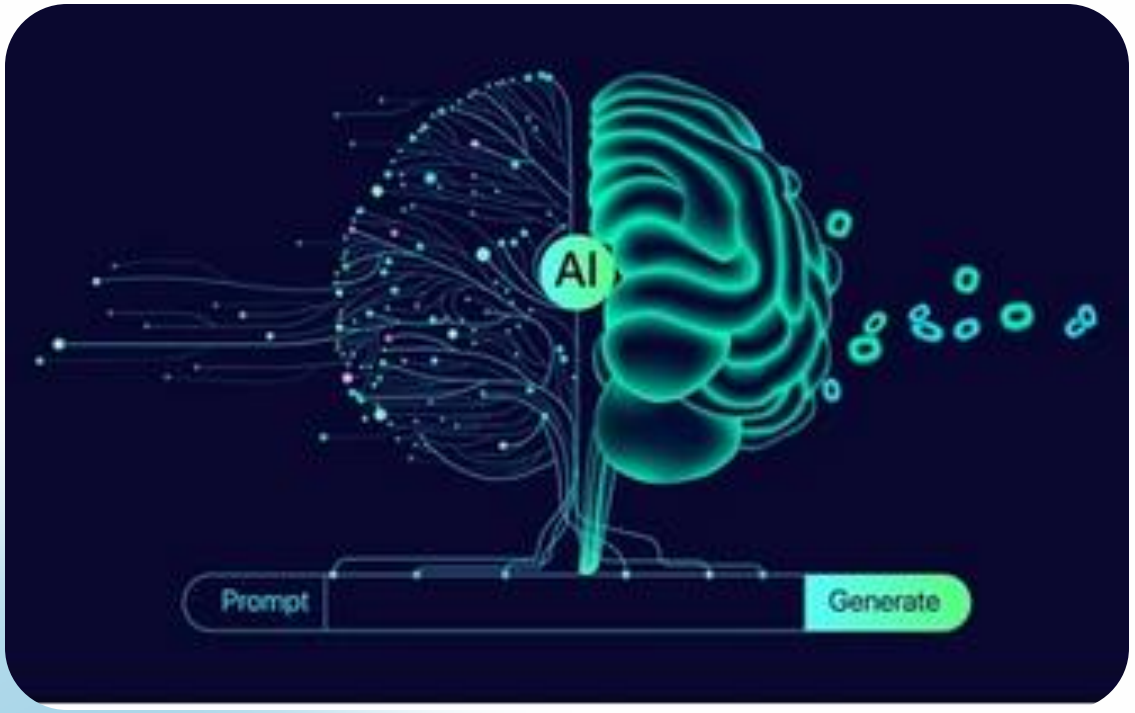
Measurement: Ensures performance tracking and quantifiable value demonstration

Governance: Protects against risks while ensuring safe, ethical deployment

Autonomous agents thrive when goals align capability with human values and real-world limits.

Hierarchical v. Reactive Planning

Hierarchical v. Reactive Planning



- Two core planning approaches guide how AI agents decide and act.
- Understanding these paradigms is vital for robotics, game AI, and automated decision-making.

Understanding Hierarchical Planning

The Strategic Approach

Hierarchical planning is a top-down method that mirrors human problem-solving by breaking tasks into manageable parts.

The agent starts with a broad goal and decomposes it into detailed sub-goals, forming a structured roadmap.

This approach suits environments requiring careful preparation and systematic execution.

How Hierarchical Planning Works

1. **High-level objective definition**:

The system sets the primary goal, forming the basis for all planning.

2. **Hierarchical decomposition**

The goal is broken into layered sub-goals, each more specific and actionable.

3. **Structured execution**

The agent follows the plan step by step, completing sub-goals in sequence.

Reactive Planning: Acting in the Moment

The Adaptive Approach

The adaptive approach uses reactive planning, focusing on immediate response over long-term strategy.

The agent monitors its environment and makes real-time decisions without detailed roadmaps.

This method prioritises speed and flexibility, ideal for fast-changing, unpredictable settings.

The agent’s strength lies in rapid assessment and appropriate action, not pre-planning.

Strengths and Weaknesses Analysis

Hierarchal Planning

Strengths	Weaknesses
Strategic, big-picture thinking approach	Reduced flexibility when conditions change
Excellent for complex, long-term projects	Time-intensive planning phase
Organised and systematic execution	Potential for analysis paralysis
Optimal resource allocation	

Strengths and Weaknesses Analysis

Reactive Planning

Strengths	Weaknesses
Fast response to changing conditions	Can be short-sighted in approach
Highly adaptable and flexible	May miss long-term optimisation opportunities
Minimal upfront planning required	Less efficient overall resource usage
Excellent in dynamic environments	

Real-World Applications

Wedding Planning: Hierarchical Success

Rigid structure limits flexibility when last-minute issues arise

- Book venue → Hire catering → Send invitations → Arrange entertainment
- Planned months in advance with detailed schedules

Real-World Applications

Traffic Navigation: Reactive Mastery

- Driving in rush hour without GPS shows reactive planning: the driver responds instantly to roadblocks and congestion.
- This real-time adaptation keeps traffic moving, though the route may be less efficient.

Key Takeaways for AI Planning

Choose based on environment

- Use hierarchical planning in stable, predictable settings where long-term optimisation is key.
- Apply reactive planning in dynamic, unpredictable situations needing immediate response.

Hybrid approaches work best

- Modern AI combines hierarchical planning for strategy with reactive elements for tactical adjustments

Context determines success

- No single approach is superior; effectiveness depends on the problem's specific needs and constraints.

Planning Strategies in Action

Understanding when to use structured roadmaps versus real-time responses

Two Fundamentally Different Approaches

Hierarchical planning

The Strategist's approach

Breaking down big goals into structured sub-goals with clear dependencies and timelines

Reactive planning

The Firefighter's method

Making split-second decisions based on real-time environmental signals and immediate conditions

Each approach suits different contexts; success depends on choosing the right strategy.

Hierarchical Planning in Practice

Family Vacation to Italy: A Structured Approach

Book international flights:
Secure round-trip tickets early for better prices and availability

Reserve hotel accommodations:
Confirm hotels in Rome, Florence, and Venice with travel links.

Create daily itinerary:
Plan activities: Day 1 Rome (Colosseum), Day 2 Florence (Uffizi), Day 3 Venice (St. Mark's)

Purchase museum tickets:
Buy advance tickets to skip queues and guarantee entry

When Structure Meets Reality

The Challenge of Rigid Planning

- Hierarchical planning builds tightly structured schedules with interdependent steps.
- **Risk factor:** A single disruption, like a cancelled train to Florence, can trigger a domino effect across the entire itinerary.
- This approach offers strong optimisation but is highly vulnerable to unexpected changes.

Reactive Planning in Action

Emergency Response: Split–Second Decision Making

Immediate response: Block hallway to prevent spread without hesitation or consultation.

Instant adaptation: Abandon current task and switch to life-saving rescue operations.

Real-time adjustment: Reroute entire team based on new conditions without delay.

Environmental signal: Smoke detected

Priority shift: Person trapped

Unexpected development: Fire spreads

The Power of Real-Time Response

The firefighter relies on immediate signals and current conditions, not long-term planning.

- **No predetermined map:** Success relies on situational awareness, not fixed plans.
- **Environmental awareness:** Real-time monitoring guides every decision.
- **Adaptive excellence:** Effectiveness stems from flexibility and instant response.

Side-by-Side Comparison

Feature	Hierarchical planning 🗂️	Reactive planning ⚡
Approach	Structured roadmap with sub-goals and dependencies	Direct response to current environmental conditions
Time horizon	Long-term strategic focus	Short-term tactical responses
Flexibility	Less flexible; disrupted by changes	Highly flexible; adapts instantly
Efficiency	Optimal in stable environments	Effective in dynamic uncertainty
Best use case	Family vacation planning	Emergency response situations

Choose Your Strategy

The Strategist

Hierarchical planning excels in stable, predictable environments where long-term optimisation matters most.

The Firefighter

Reactive planning dominates in unpredictable, dynamic situations requiring immediate adaptation.

Success comes from matching your planning approach to your environment's characteristics.

Autonomous Agents: Structured Long-Term Planning vs. On-the-Fly Reactions

Exploring how AI agents balance strategic foresight with instant adaptation to create truly autonomous systems

Structured Long-Term Planning

Task decomposition

Breaks complex goals into manageable subgoals and sequences actions ahead of time

Classical planning

Uses hierarchical frameworks, symbolic AI, and classical planners like PDDL for systematic execution

Strategic foresight

Enables error recovery and multi-step workflows with predictive capabilities

Structured Long-Term Planning

Real-World Analogy

Like a chess grandmaster, planning several moves ahead, anticipating responses and preparing counters before the first move.

On-the-Fly Responses

Instant decision-making

Makes decisions moment-to-moment based on immediate inputs and current environmental context

Adaptive learning systems

Employs reactive loops, reinforcement learning, and fast heuristic reasoning for rapid adaptation

Dynamic environment mastery

Excels in unpredictable, rapidly changing environments requiring immediate response and quick pivoting

On-the-Fly Responses

Real World Analogy

Like a firefighter in a burning building, reacting instantly to smoke, structural shifts and emergencies without a preset plan.

Hybrid Agents

Combining Planning and Reaction

Modern autonomous agents blend both approaches for optimal robustness and adaptability in complex settings.

1. Strategic foundation

Establish long-term goals and structured pathways using planning frameworks

2. Real-time adaptation

Continuously monitor environment and adjust actions based on immediate feedback

3. Dynamic integration

Use frameworks like ReAct prompting to interleave reasoning and acting fluidly

Hybrid Agents

Prime Example

Self-driving cars plan efficient routes using traffic data, while reacting in real time to pedestrians, obstacles and road changes.

Key Components Enabling These Modes

Vector databases support stateful understanding, preserving short-term context and long-term experience for informed decisions.

Advanced Large Language Models like GPT-4 and Claude enable advanced planning and contextual problem-solving.

API orchestration, autonomous scripts and self-healing systems ensure smooth execution with error recovery and adaptability.

Perception & memory

Reasoning engines

Tool use & execution

Real-Time Training Analogy

Learning to Drive

Pre-drive planning
Review route, rules, and anticipate hazards before driving

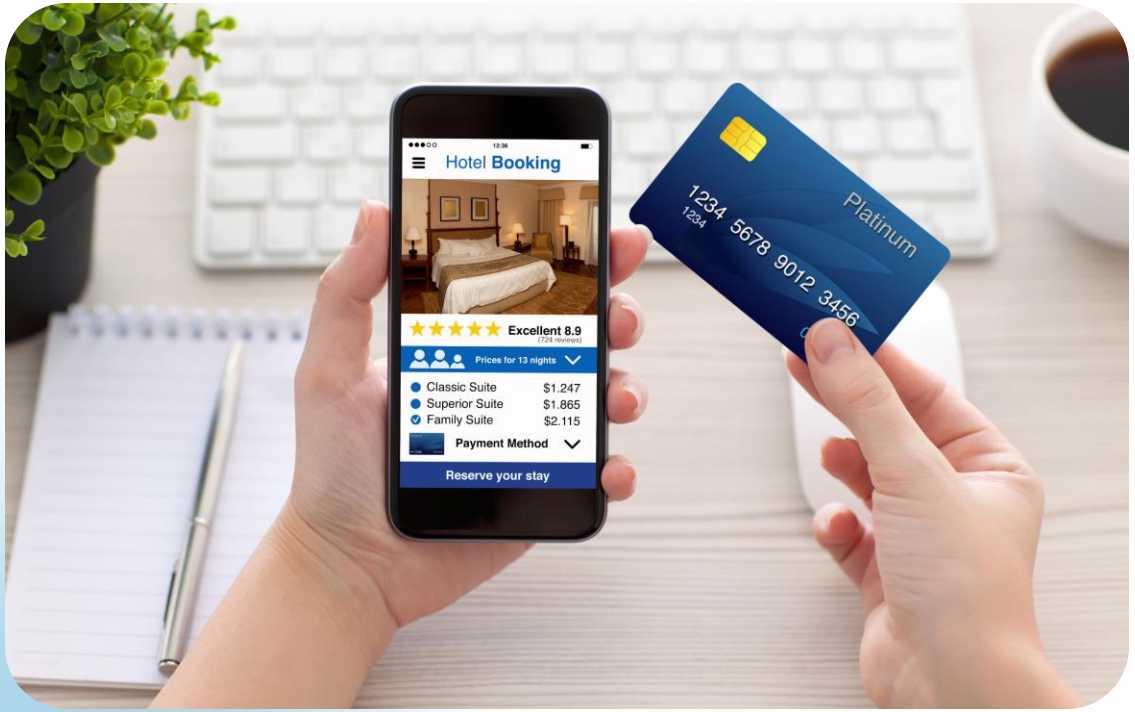
Active driving reactions
React instantly to stops, pedestrians, weather and road changes

Continuous learning
Refine driving style through experience, feedback and evolving conditions

Training Insight: Like human drivers, autonomous agents learn through structured prep, real-time experience and ongoing feedback loops.

Case Study

AI Travel Assistant Booking a Trip



- **Strategic planning phase**: Research destinations, compare flights, check hotel ratings and build a full itinerary with backups and budget in mind.
- **Real-time adaptation**: Track flight delays, weather and availability; adjust bookings and alert travellers instantly
- **Seamless integration**: Balances long-term planning with reactive updates for smooth travel despite disruptions

Why This Matters: Business & Tech Impact

Ideal for complex workflows like supply chain, finance and legal tasks where systematic methods deliver superior results

Suitable for fast-paced domains like robotics, live support and gaming where split-second choices drive success

Achieves 10–20× productivity gains through continuous learning and adaptive optimisation across sectors

Structured planning excellence

Reactive agent dominance

Hybrid agent revolution

The Future of Autonomous Agents

Perfect balance

Blending structured planning with agile reactions enables autonomous systems that think like humans.

Success framework

Memory, reasoning, tool use and self-reflection underpin effective agent design.

Industry transformation

AI agents will reshape work by combining strategic foresight with instant adaptability

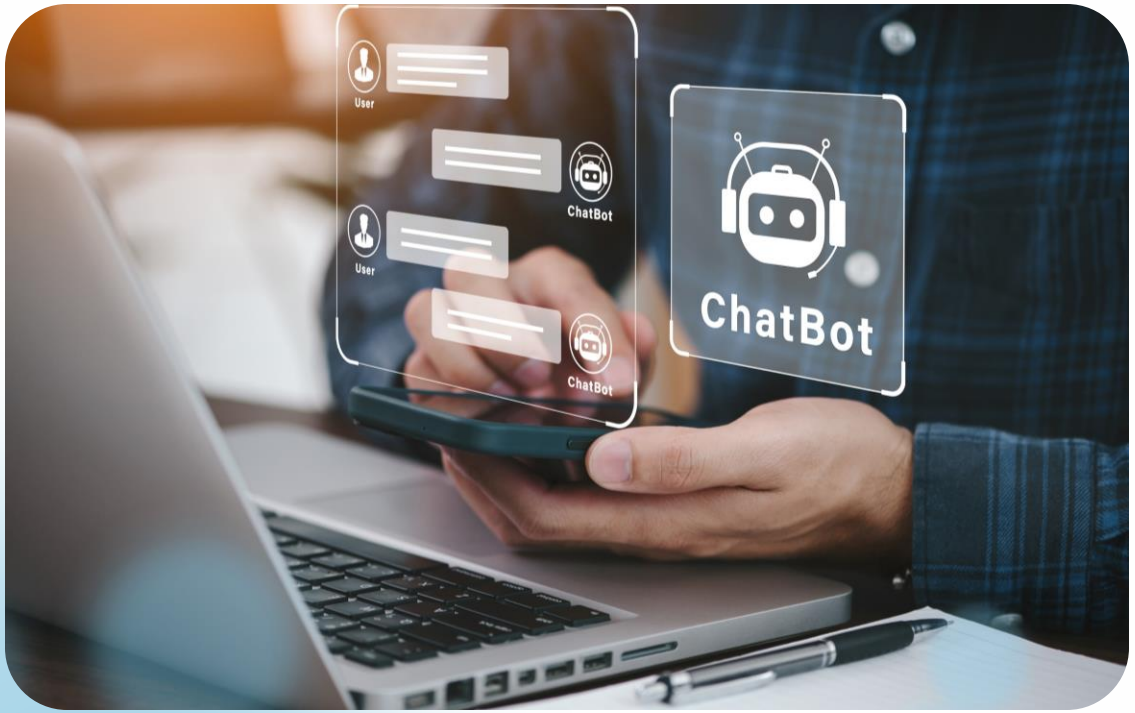
Ready to harness autonomous agents?

Begin with clear goals, solid planning and real-time feedback for ongoing improvement.

Multi-Agent Planning Strategies

Coordinated decisions across teams of autonomous agents

What Are Multi-Agent Planning Strategies



- Multi-agent planning lets AI, robots and software coordinate seamlessly toward shared or individual goals.
- Agents collaborate rather than operate in isolation, avoiding inefficiencies and overlaps.
- The shift from solo decision-making to joint planning unlocks emergent intelligence.
- Teams of agents solve complex, multifaceted problems beyond the reach of any single system.

Why Multi-Agent Planning Matters

Complex problem scale

Challenges like supply chains optimisation, autonomous vehicles coordination and distributed computing demand multi-agent solutions for scalable problem-solving.

Conflict prevention

Coordination avoids resource clashes and operational inefficiencies—agents collaborate to boost throughput and cut waste.

Ecosystem reality

Modern AI favours agent ecosystems over monolithic models; the future lies in collaborative intelligence.

Planning Strategy Types Overview

Centralised

Single coordinator manages all agents

Decentralised

Agents make local decisions with shared updates

Cooperative

Agents actively collaborate toward shared goals

Competitive

Agents balance individual goals with system rules

Centralised v. Decentralised Planning

Centralised planning

How It works: A central coordinator directs all agents, ensuring unified control and conflict-free operation.

- **Pros:** Efficient coordination, no duplicate effort and optimal resource use.
- **Cons:** Single point of failure, limited scalability and risk of bottlenecks

Example
Air traffic control systems, where one tower manages all aircraft in a region.

Decentralised planning

How it works: Each agent acts independently using local data, while sharing updates and coordinating via distributed communication.

- **Pros**: Scalable, fault-tolerant and adaptive to local conditions.
- **Cons**: Risk of conflict without clear communication; complex coordination.

Example
Autonomous vehicles using vehicle-to-vehicle (V2V) communication and shared sensor data.

Cooperative and Competitive Planning Models

Cooperative planning

Agents align on shared goals, support one another and optimise collective performance through collaboration and resource sharing.

Real-world example:
Warehouse robots coordinate tasks—retrieval, transport, sorting—for efficient division of labour.

Competitive/Mixed planning

Agents pursue individual or conflicting goals within shared rules, balancing self-interest with system-wide coordination.

Real-world example:
Trading bots compete for profits while respecting market protocols and maintaining stability..

Understanding Through Real-World Analogies

Orchestra conductor

Centralised planning:
A single coordinator directs all agents, ensuring unified timing, coordination and harmonic execution.

Soccer team

Decentralised planning:
Players make individual decisions based on field conditions while coordinating through shared strategy and real-time awareness.

Understanding Through Real-World Analogies

Restaurant Kitchen

Cooperative planning:
Chef, sous-chef and waitstaff perform specialised roles that integrate seamlessly to deliver exceptional service.

Traffic Intersection

Competitive planning:
Drivers pursue individual goals while coordinating through traffic rules to maintain collective flow.

Professional Applications in Industry

Autonomous fleet management

- Agents coordinate drone, vehicle, and robot fleets to optimise routes and avoid collisions.
- Real-time and predictive planning boosts delivery efficiency and operational safety.

Smart grid optimisation

- Energy providers, storage systems and predictors balance supply and demand dynamically.
- Agents integrate renewables, manage load distribution and maintain grid stability.

Professional Applications in Industry

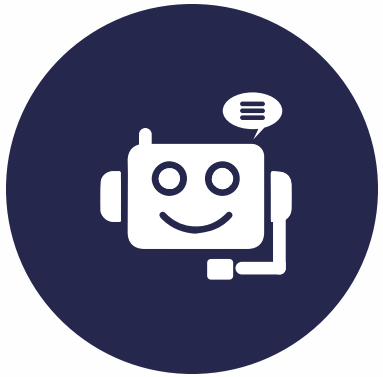
Financial market systems:

- Trading agents compete to optimise individual portfolios while preserving market liquidity.
- High-frequency bots coordinate to prevent manipulation and uphold system stability.

Healthcare resource coordination:

- AI systems manage staff schedules, equipment use and patient flow across hospital operations.
- Multiple agents coordinate emergency response and optimise resource allocation.

Success Factors for Multi-Agent Systems



Communication

Clear channels, standard formats and real-time updates enable agents to coordinate effectively.

Goal alignment

Shared objectives, metrics and incentives align agent actions and prevent conflict.

Conflict resolution

Built-in mechanisms resolve disputes and coordination failures, ensuring resilient operation.

Key Takeaway

Multi-agent planning turns AI from isolated performers into coordinated collaborators.

Coordination + decision-making at scale

AI's future lies in systems that coordinate complex decisions across multiple agents—solving problems beyond any single system's reach.

Success through strategic design

Effective multi-agent systems need well-designed communication, goal alignment and conflict resolution for optimal coordination.

Reinforcement Loops in AI Planning

Feedback-driven adaptation and iterative optimisation for autonomous systems that learn from experience and continuously improve their decision-making capabilities.

What Are Reinforcement Loops

- Reinforcement loops shift planning from static decisions to dynamic, adaptive systems.
- Intelligent agents use continuous feedback to learn and optimise strategies over time.
- Rigid algorithms become flexible, self-improving systems responsive to change.
- This forms a strong foundation for autonomous AI capable of handling uncertainty and complexity.

The Three-Step Foundation

Act: Agent executes actions based on current strategy and available information.

Adapt: Agent refines its strategy using learned experience to improve future decisions.

Observe: System receives environmental feedback—success, failure, rewards or penalties.

This continuous cycle transforms basic reactions into intelligent, adaptive behaviour for complex, changing environments.

Why Reinforcement Loops Matter

Dynamic environments

Adaptive systems adjust continuously to changing conditions, unlike static plans which quickly become obsolete.

Resilience & recovery

Reinforcement loops help systems recover from failure by learning and generating alternative strategies.

Long-term optimisation

Each cycle improves performance, evolving trial-and-error into refined decision-making.

The Reinforcement Learning Connection

Foundation of Autonomous Systems

- Reinforcement loops let agents learn through experience, not programming.
- This mirrors biological learning—trial, error and gradual improvement.
- AI adapts to new situations without human input, essential for robotics, game AI and autonomous vehicles.
- Modern RL achieves superhuman performance in tasks like chess, Go and strategy games.

Learning Through Experience

Real-World Analogies

Athletic skill development

A basketball player improves free throws through repeated feedback-driven practice until muscle memory forms.

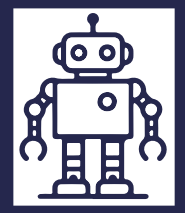
Intelligent navigation

Google Maps refines routes using real-time traffic feedback to optimise future journeys.

Culinary mastery

A chef perfects recipes through tasting and adjustment, turning feedback into refined techniques and signature dishes.

Enterprise AI Applications



Robotics & manufacturing

- Robots learn precise force control through iterative assembly tasks.
- They balance speed and sensitivity to prevent damage while maintaining efficiency

Recommendation engines

- Platforms like Netflix and Spotify adapt suggestions using user feedback.
- Every interaction refines recommendations for individuals and similar users.

Enterprise AI Applications

Autonomous vehicles

- Self-driving cars adjust driving strategies based on real-time conditions.
- Each mile driven improves safety and navigation efficiency.

Supply chain optimisation

- Logistics systems adapt inventory, routing, and operations using performance data.
- Continuous feedback reduces waste and boosts overall efficiency.

The Detailed Reinforcement Process

Initial planning phase

Agent assesses current state and options, selecting actions using knowledge, heuristics or exploration.

Action execution

System applies the chosen strategy, committing resources and influencing variables or stakeholders.

Feedback collection

Environment returns measurable outcomes—performance, errors, satisfaction, efficiency or penalties.

Strategy refinement

Agent uses feedback to update models, adjust parameters and improve future planning.

Iterative improvement

Cycle repeats with refined strategies, driving increasingly effective decision-making over time.

Static Plans v. Living Processes

Without Feedback	With Feedback
Plans remain fixed regardless of outcomes	Systems continuously learn and adapt
No learning from failures or successes	Performance improves through iteration
Vulnerable to environmental changes	Resilient to changing conditions
Performance stagnates over time	Self-optimising and autonomous

Transforming AI Planning

Traditional AI relies on fixed rules and cannot adapt or learn from experience.

Reinforcement loops enable dynamic systems to evolve, adapt and optimise through real-world feedback.

From static scripts

To living processes

Reinforcement loops transform rigid automation into adaptive intelligence that learns and improves through experience.

Q & A

Thank you